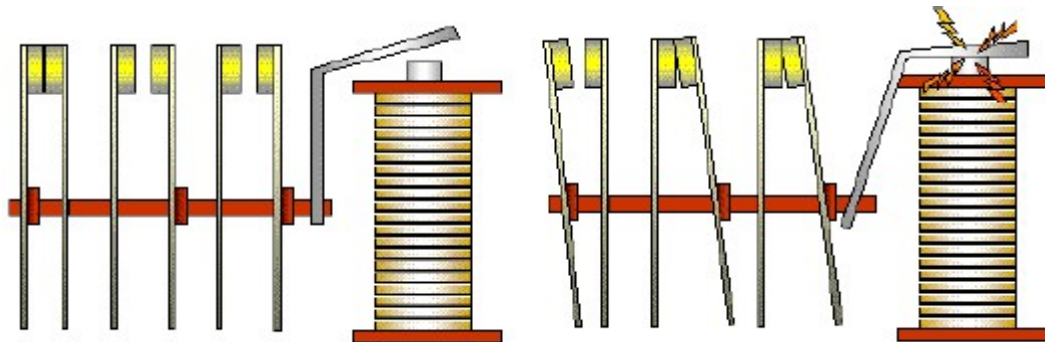


1 INTRODUCTION

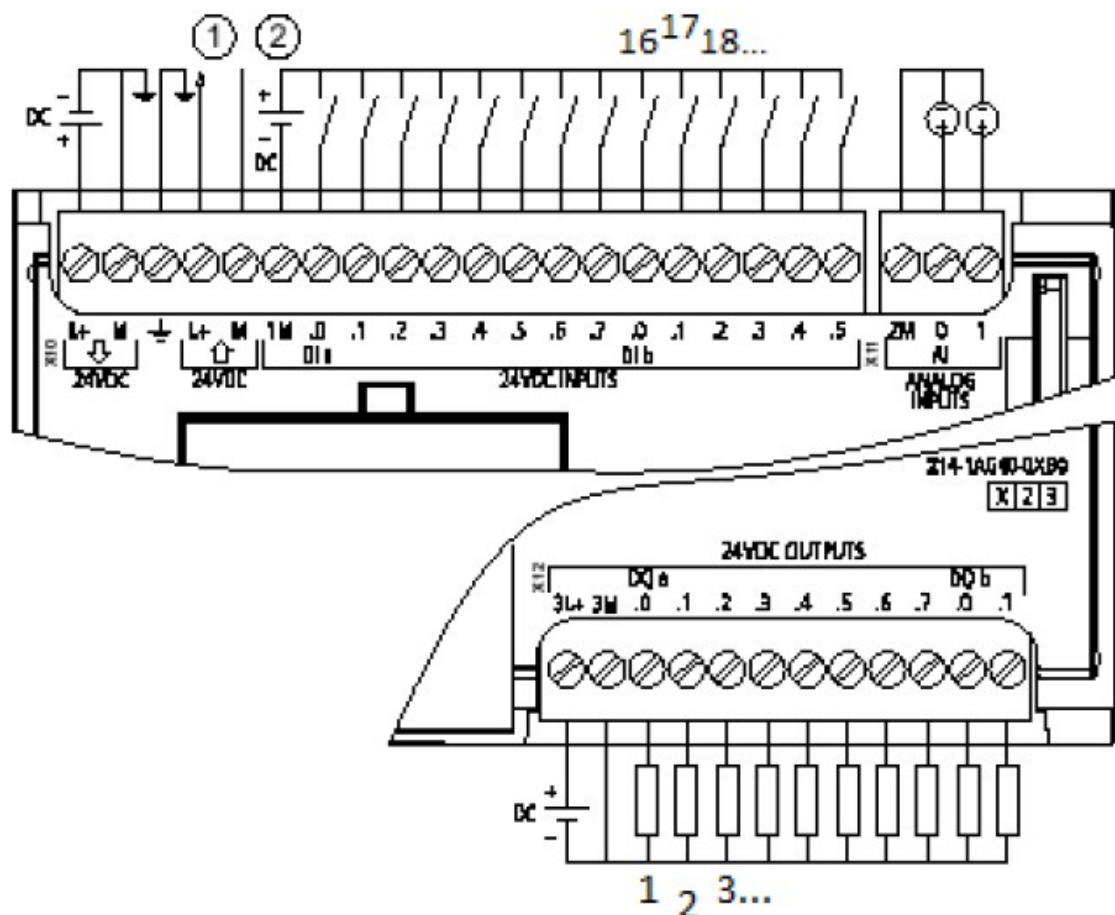
In this work, a logic application (hw configuration+program) is made that controls the motor through a relay. The relay will have to be used because the current drawn by the motor would overload the logic binary output. A relay is a type of electromechanical switch that can be controlled on/off at a relatively low current (power) but can control even a large current with its contacts. Figure 1 shows the principle of operation of the relay.



When the coil receives voltage from the outside (pictured right), a magnetic field is created that mechanically switches the state of the contacts. The contacts that were open close and the contacts that were closed open. Through these contacts goes the current of the controlled device. That is, the device controlled by the state of the contacts is either in operation or is not. The power passing through the contacts is usually significantly higher than what could be fed through logic. In addition, the voltage is typically second. Thus, the coil circuit and the contacts are not electrically connected in any way to each other, but are galvanically separated. The only connection between the guiding power and the controlled power is mechanical. When the relay coil is energized, it is said that the relay is pulled. When the coil is de-energized, it is said to have been discharged.

MOTOR CONTROL, FAN

When the relay is connected to the binary output of programmable logic, the output of the logic (eg. Q124.0 true or false) interrupts the voltage in that channel and thus the relay either pulls or emits. Figure 2 shows one example of connecting a relay to logic.



At the top, the connection of inputs with logic (14 separate binary inputs) is shown. In them, a voltage of +24 VDC is applied to the field (for example, to the sensor). If the contact in the field is closed, the voltage can be used on a channel-by-channel basis to tell the logic whether the tip connected to that channel was open or closed (voltage at the logic connector indicates that the contact, or tip, was closed).

MOTOR CONTROL, FAN

As for the outputs that appear at the bottom (10 pieces of binary outputs), the operation, of course, is slightly different. In them, the supply voltage is connected to the card terminal 3L+, the division of which by channels is handled inside the board. When the output is turned on, the corresponding output terminal is energized. At other times, of course, the output terminal is de-energized. When a relay coil is connected to the terminal, the relay in question pulls and connects the device connected to it to the voltage (usually 230 / 400 VAC). The relay and contactor coil connectors are typically marked A1 and A2. A1 is the terminal to which the voltage is connected and zero is connected to A2.

2 INFORMATION ABOUT THE WORKSTATION

The programmable logic at the base is a Siemens S7 1214C. It has an integrated Profinet interface, as well as binary inputs and outputs. The motor is controlled by a relay, the traction coil of which is connected to the binary output of logic. The status of the relay (dragging or letting) is read to the logic so that in the program it can be concluded whether the control is successful or not.

3 SAFETY

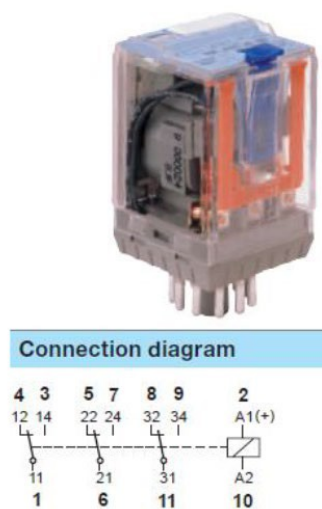
Care and care should be taken when working with electrical appliances at the workstation. In any case, therefore, it is not allowed to go into connections other than 24 VDC. In addition, one should beware of the rotating parts of the equipment.

4 TASK

1. Open TIA Portal V16 and create a new project. Please add the CPU as a new device. Pay attention to the correct CPU type (the exact type number can be found in small print on the CPU). Please ask for help with this section if necessary.

MOTOR CONTROL, FAN

2. At this point, keep the logic plug out of the socket (the logic will certainly be de-energized). Find out for yourself how the connections between the logic and the group of terminal blocks on its pedestal are implemented. Do not make any changes to them. How is the voltage from the source connected above the CPU branched and from which terminal blocks are 24 VDC and 0 VDC voltages available? Are the voltage supplies for the cards already ready (cf. Figure 2)?
3. (Still with the logic plug disconnected from the outlet.) Make the necessary connections for the work. Thus, the intention is to make connections only to the terminal blocks of the logic stand and from them to the "field", that is, towards the motor. Connect the relay coil to the output channel (Q0.0) and the contact to the input (I1.0). Find out the correct connector numbers using Figures 2 and 3. The addresses depend on the hardware settings and it is worth making sure that for digital outputs and inputs 'I address' 0... 1 and 'Q address' 0... 1 are also addressed in the program. When making connections, the principle images on the back side of the logic cover still help. Figure 3 shows the terminal numbers of the relay to be used. The relay and its base have different numbers, both of which are shown in the attached figure.



Present the connections you have made to the instructor before connecting the electricity to the logic!

MOTOR CONTROL, FAN

4. Make a function (FC1) for which you are making the application. Make a program for FC1 that implements the functions described below (the I/O address table is given in Appendix 1, load the program into logic and test the operation of the program in monitoring mode as necessary download instructions on the following page):

- The engine starts from the rising edge of the Start switch (I0.0) and stops at the Stop switch (I0.1). The Start switch is the leftmost switch.
- At the beginning of the start-up attempt, a one-second monitoring period is triggered. If the relay controlling the engine does not pull during this time (i.e. the feedback is not reached (I1.0)) the engine control must stop and the error light must come on (e.g. to an unconnected output of the CPU). The alarm must also be generated if the logic receives feedback information even though the motor should be at a standstill (e.g. a contact welded closed). Thereafter, it shall not be possible to start the motor until the error has been acknowledged by the ascending edge of the acknowledgement switch (I0.6). The error light turns off when the error has been acknowledged and after this it is possible to start the engine again.
- The restart of the motor is allowed after stopping only after the motor has been at a standstill for 5 seconds. Avoid unnecessary start-ups!

5. When the application is ready, load it into logic and test. You can transfer the program (OB1 and FC1) using the method you learned in the class, or alternatively in the following way: activate OB1 and FC1 → press the right mouse button on them to open the menu → select the command SOFTWARE found in the submenu under DOWNLOAD TO DEVICE from the menu. After you have tested the program, present it to the teacher.

ADDITIONAL TASK: Make counters for the number of motor starts and alarms. The counters shall be resettable with the input listed in the Annex.

MOTOR CONTROL, FAN**5 ATTACHMENTS****Annex 1. I/O's addresses**

ADDRESS	EXPLANATION
I0.0	"Start" switch input (rising edge)
I0.1	"Stop" switch input
I0.7	Alarm acknowledgment (acknowledgment with rising edge)
I1.0	Feedback from contactor
I0.6	Counter reset switch input
Q0.0	Output for motor
Q0.7	Alarm light

MOTOR CONTROL, FAN